An open dataset on Magic The Gathering Cards for teaching biostatistics.

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in prep for Journal of Statistics and Data Science Education

fit

JSDSE has a specific article category called "Datasets and Stories", designed for papers introducing interesting multivariate datasets for teaching statistics, including learning outcomes, potential pitfalls, and classroom applications. It also explicitly covers statistics education in the health sciences.

dataset Zuliani, Mario; lortie, cj (2020). Magic The Gathering Data. figshare. Dataset. https://doi.org/10.6084/m9.figshare.12797474.v2 (need to update with new file in email)

7,713 Card observations 763 Unique card names

3 MTG sets

540 Pack sampling units

4 Rarity classes

prices in 2021 and now 2026

Table 1. A list of teaching implementations using MTG card data

Research question	Statistical tests and analyses	Learning outcomes for biostatistics
Are rare cards really rare?	Frequency distributions; chi-square goodness-of-fit; chi-square test of independence; 95% confidence intervals	Probability; observed vs. expected frequencies; categorical data; hypothesis testing; sampling uncertainty.
Are some card colors more powerful than others?	ANOVA; ANCOVA (controlling for mana cost and set); Tukey pairwise comparisons	Differences among groups; confounding variables; adjusted means; correlation vs. causation.
Do MTG sets differ in biodiversity?	Species richness; Shannon and Simpson diversity; Hill numbers (q = 0, 1, 2); rarefaction; species accumulation curves	Biodiversity measurement; richness vs. evenness; sampling effort; standardization; ecological sampling theory.

sample datatable

observer	set	rep	pack_card	card_name	rarity	card_type	creature_type	power	toughness	color	colorCMC	finish	n_keywords	keywords	price	price_2026
mz	Zendikar Rising	1	8	Acquisitions Expert	Uncommon	Creature	Human Rogue	1	2	1	Black	2	Regular	0	NA	1	0.25	0.23
mz	Zendikar Rising	2	17	Acquisitions Expert	Uncommon	Creature	Human Rogue	1	2	1	Black	2	Regular	0	NA	1	0.25	0.23
mz	Zendikar Rising	3	6	Acquisitions Expert	Uncommon	Creature	Human Rogue	1	2	1	Black	2	Regular	0	NA	1	0.25	0.23
mz	Zendikar Rising	3	17	Acquisitions Expert	Uncommon	Creature	Human Rogue	1	2	1	Black	2	Regular	0	NA	1	0.25	0.23
mz	Zendikar Rising	4	13	Acquisitions Expert	Uncommon	Creature	Human Rogue	1	2	1	Black	2	Regular	0	NA	1	0.25	0.23
mz	Zendikar Rising	4	28	Acquisitions Expert	Uncommon	Creature	Human Rogue	1	2	1	Black	2	Regular	0	NA	1	0.25	0.23

Sample stats

1. Are rare cards really rare?

Yes. Mythic cards represented less than 1% of the 7,713 observations. Rarity classes differed strongly from an equal-frequency distribution ($X^2 = 9{,}391.2$, $p < 0.001$).

2. Are some card colors more powerful than others?

Yes, modestly. Creature power differed significantly among colors after controlling for mana cost (CMC) and set (ANCOVA: $F(4, 319) = 3.28$, $p = 0.0117$). Discuss power.

3. Do MTG sets differ in their biodiversity?

Yes, descriptively, but the differences depend on how biodiversity is measured. Discuss how diversity stats are complex.

Question	Finding	Statistical lesson
Are rare cards really rare?	Yes; mythics were only 0.9% of observations.	Probability and chi-square tests
Are some colors more powerful?	Yes; red and green exceeded blue after adjustment.	ANCOVA, confounding, multiple comparisons
Do sets differ in biodiversity?	Yes, in different ways depending on diversity order.	Hill numbers, rarefaction, richness vs. evenness

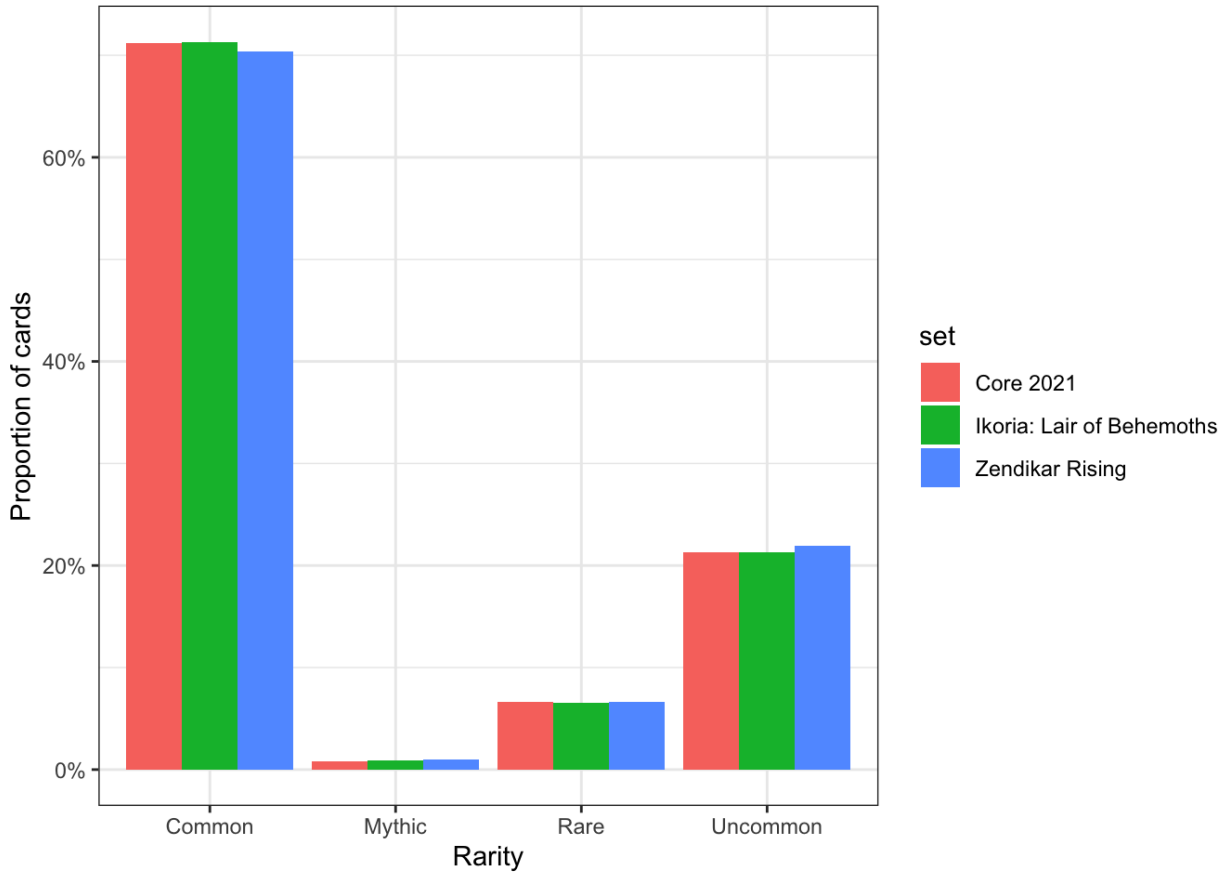


Figure 1: players

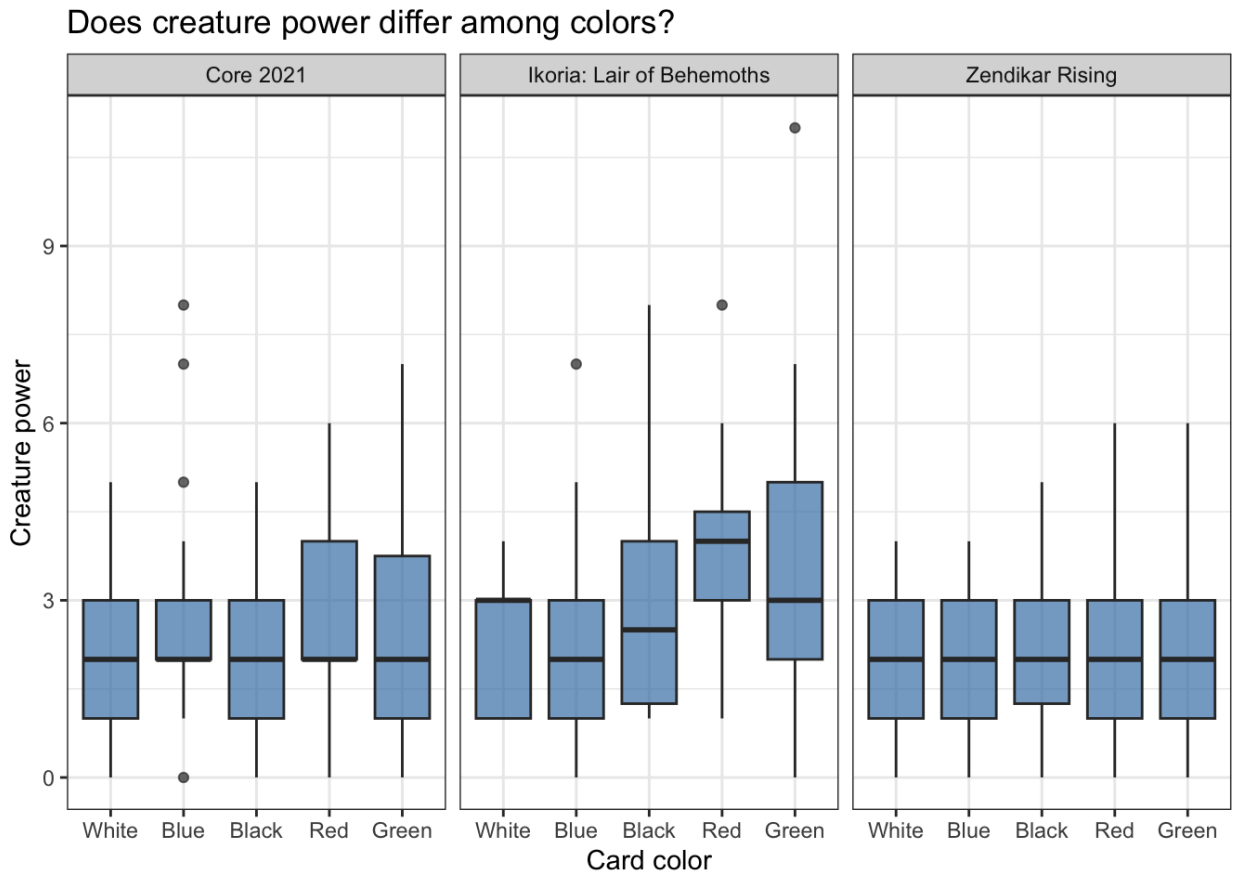


Figure 2: strategy

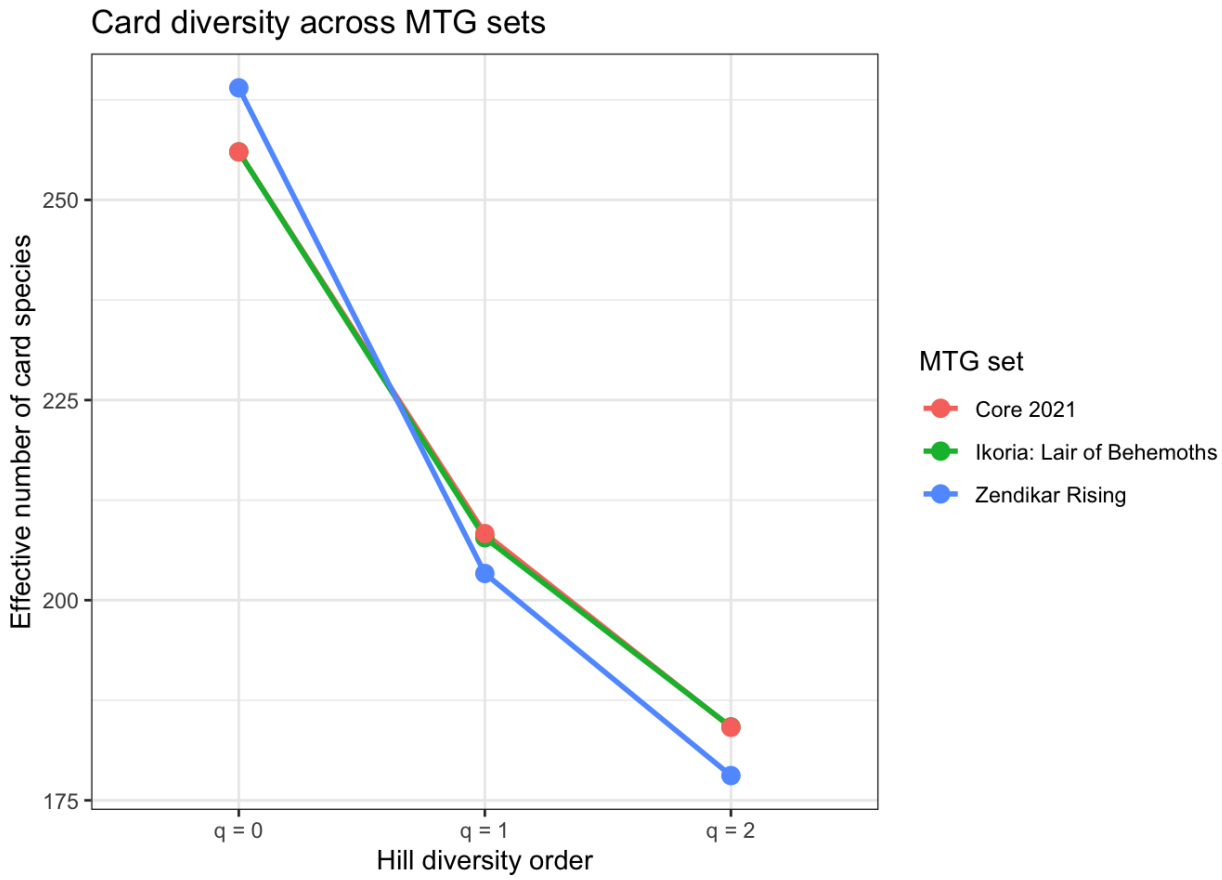


Figure 3: strategy